Intro to Mechanistic Modeling

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Learning Objectives

1. Appreciate the use of modeling in the biological sciences.
2. Differentiate between mechanistic and empirical approaches.
3. Understand the difference between stochastic and deterministic models.
4. Explain how a compartmental model functions.

Introducing Myself

My area of research is dairy cattle!

I'm also a huge nerd!

Born and raised, 1997

BS in Animal Science, 2019

Doctor of Veterinary Medicine, 2023

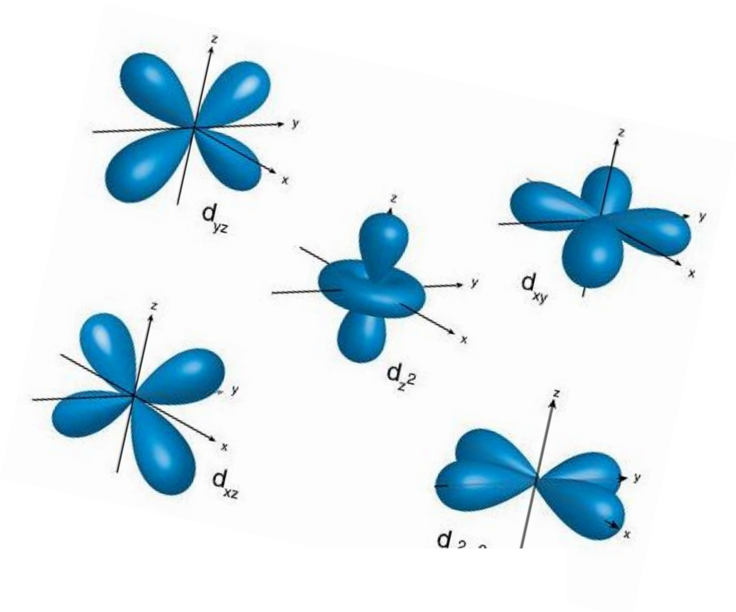
PhD in Epidemiology, exp. 2027

“All models are wrong, but some are useful.”

George Box, 1976

Modeling in Epi

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

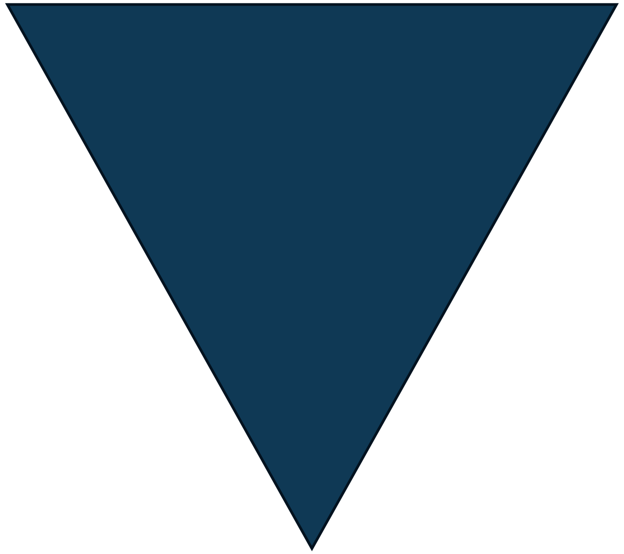


WHAT
A representation of a concept or system

WHY
To develop and test hypotheses about the interactions of components within said concept or system

HOW
Physical (rare) or abstract (common)
Data-driven or mechanistic

Data



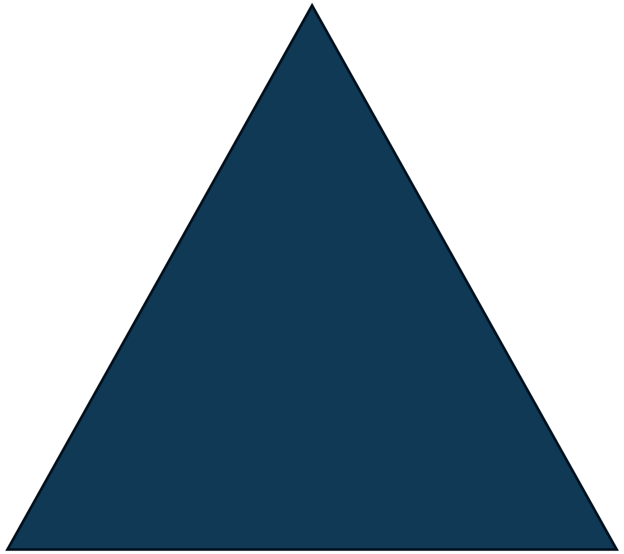
System

← Data-driven
Mechanistic →

Data-driven models fit the data, but don't explain *how/why* the data look the way they do

Mechanistic models assemble the underlying components of a system, but rely upon *assumptions* that may make their data behave *differently from reality*

Data



System

Compare Two Approaches

Say I want to model the behavior of *Listeria monocytogenes* in a dairy processing facility. My dataset includes culture results from an environmental monitoring program.

Data-driven model

I fit a logistic regression model that determines the probability that a given surface is contaminated as a function of a series of covariates (i.e., surface material, ambient temperature).

Mechanistic model

I create an agent-based model that represents the movement of individual surfaces (i.e., carts, workers, utensils) in space and the growth of the bacteria on those surfaces.

Is one model "better" than the other?

More modeling terminology

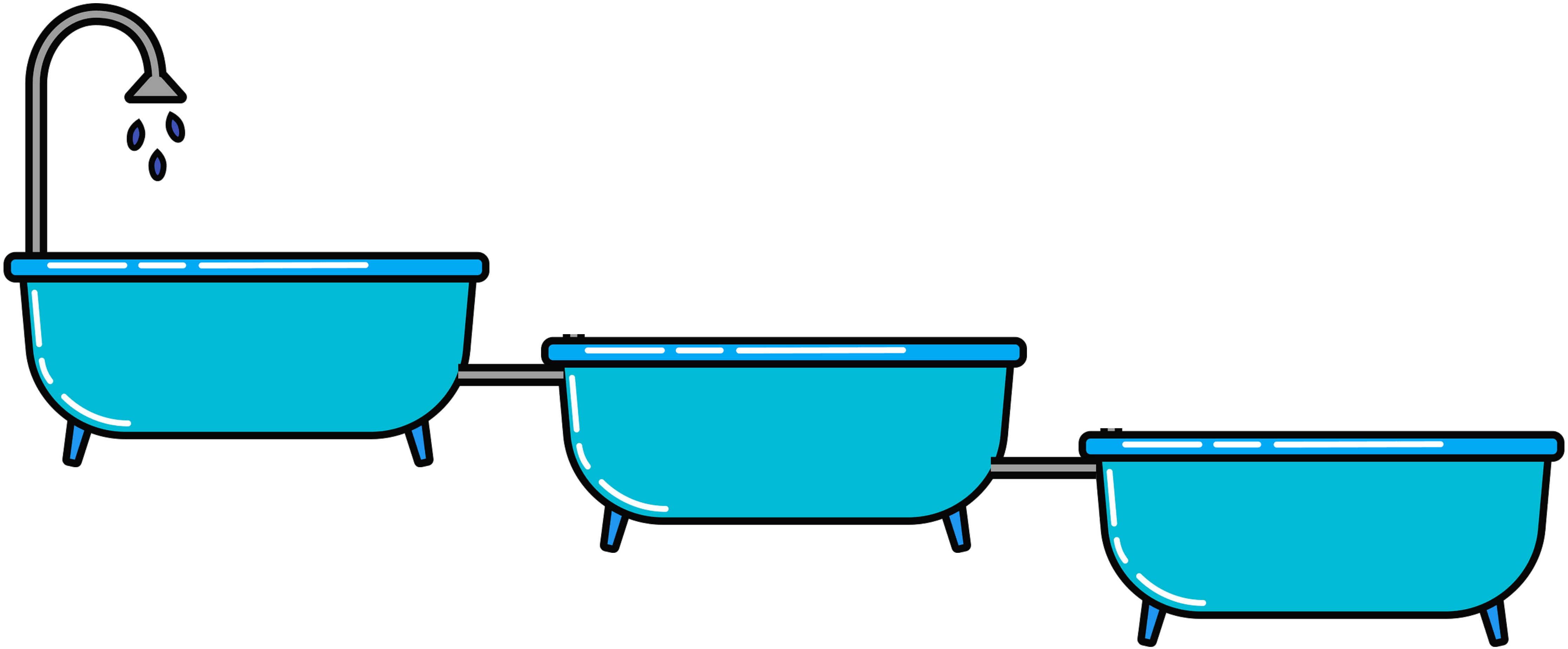
Parameter

Stochastic

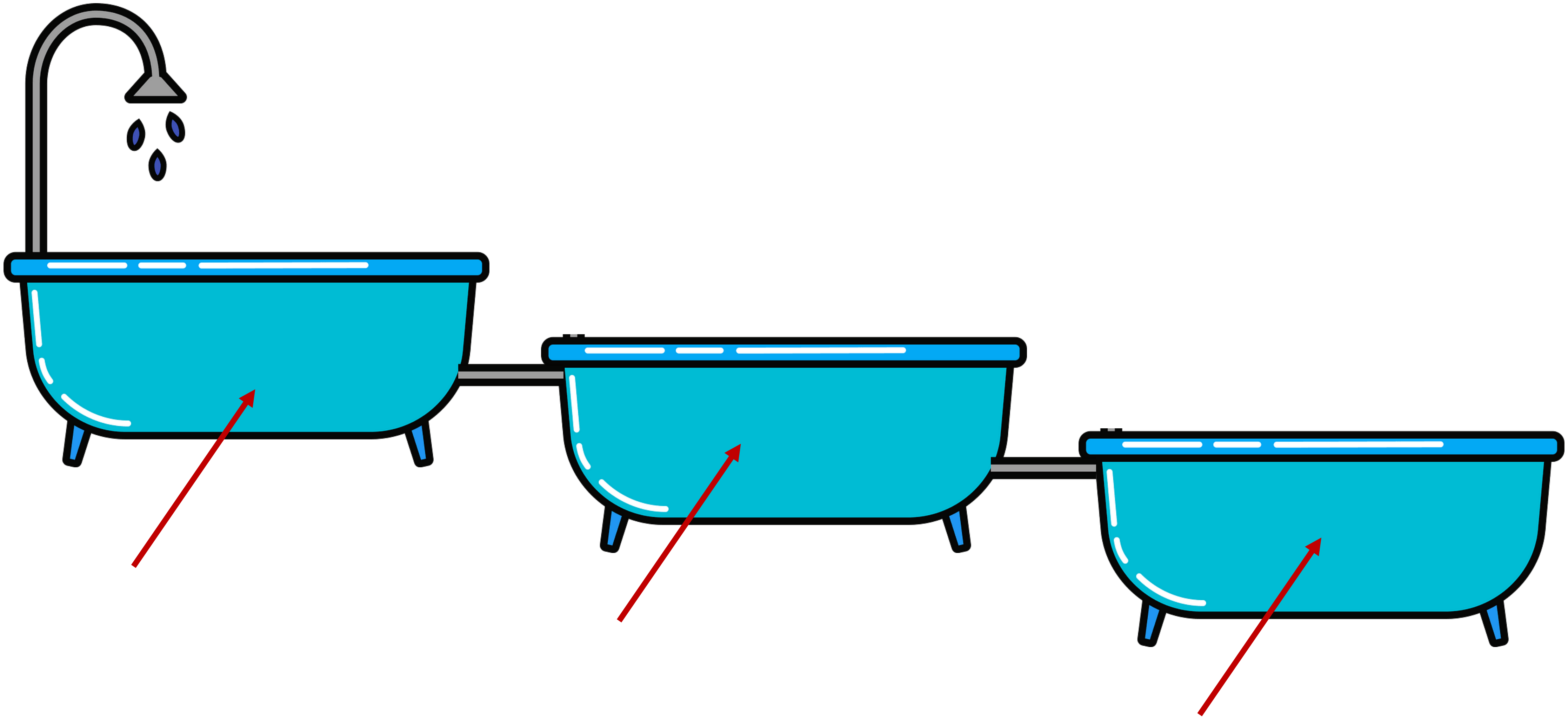
Deterministic

Pause.
Reflect.
Ask questions.

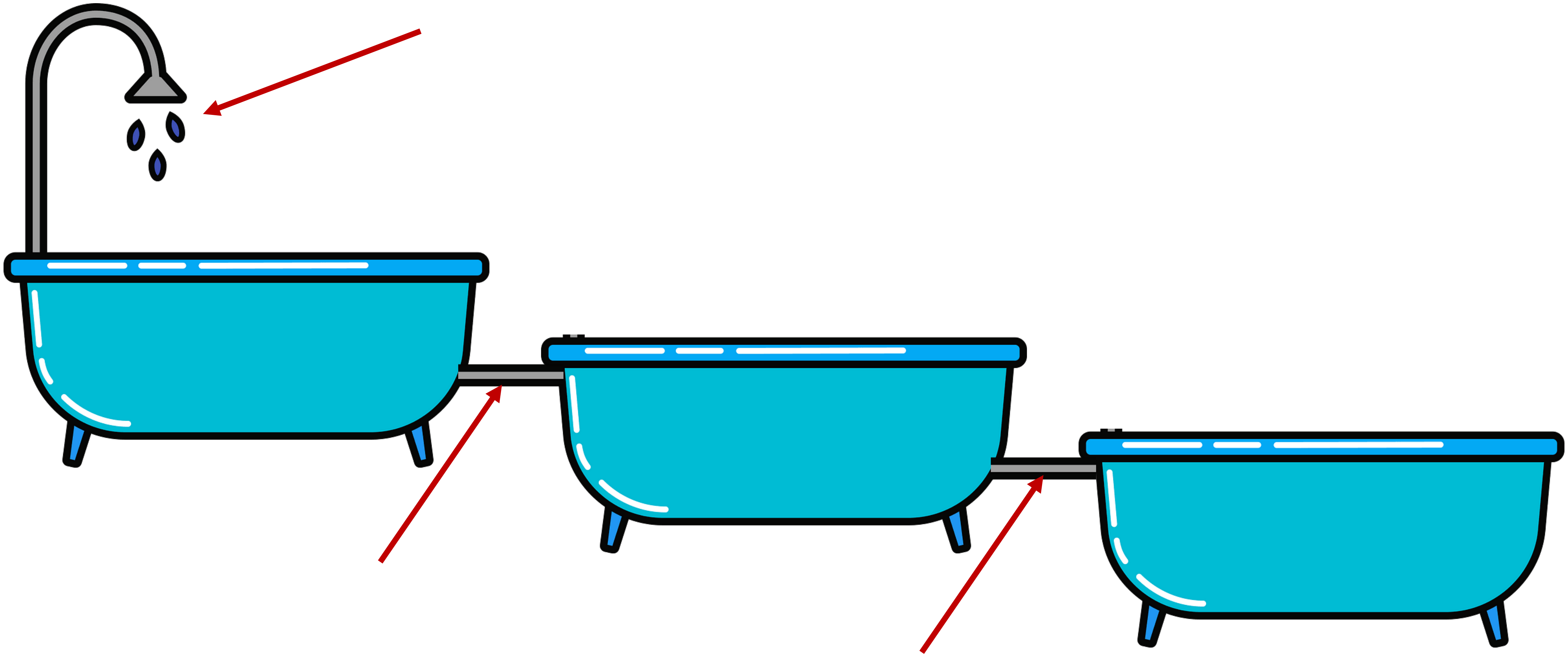
Think about a public health system or phenomenon that interests you.
How might you use modeling to study it?



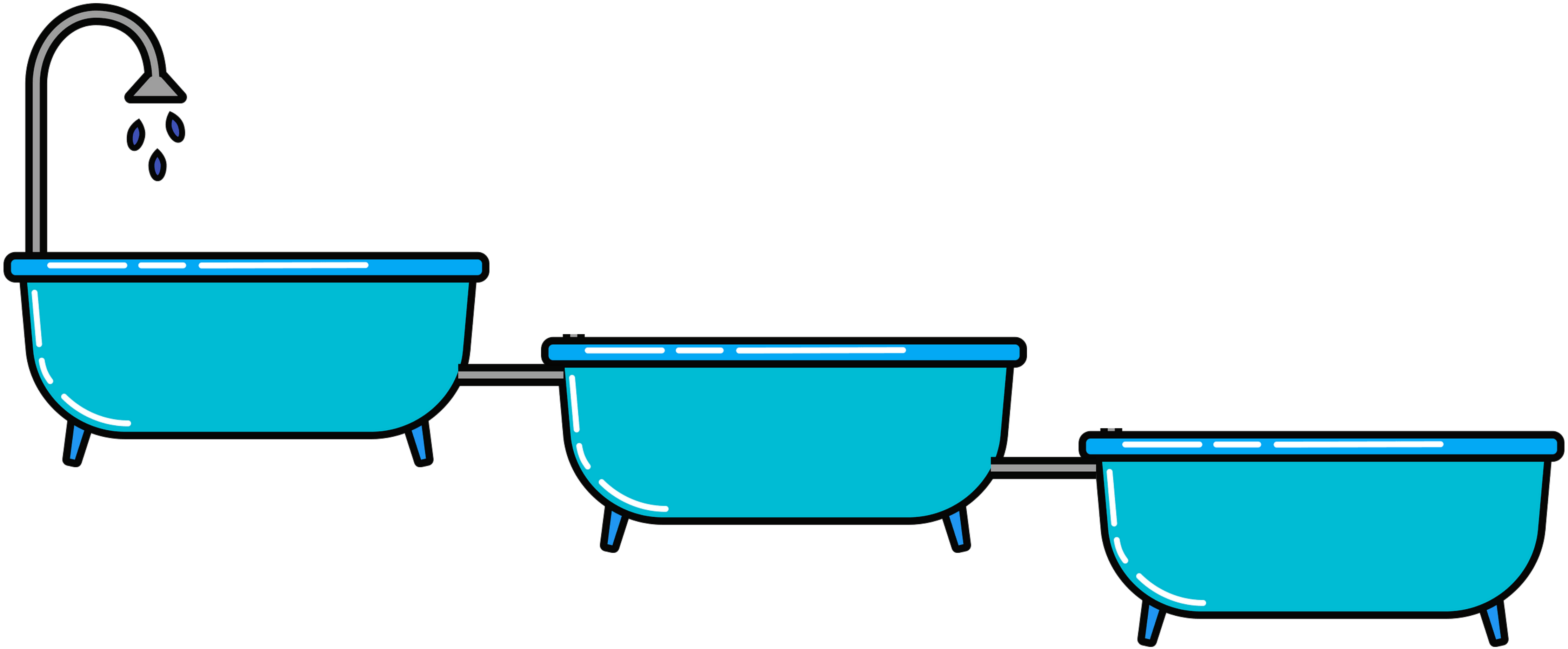
Compartmental Model



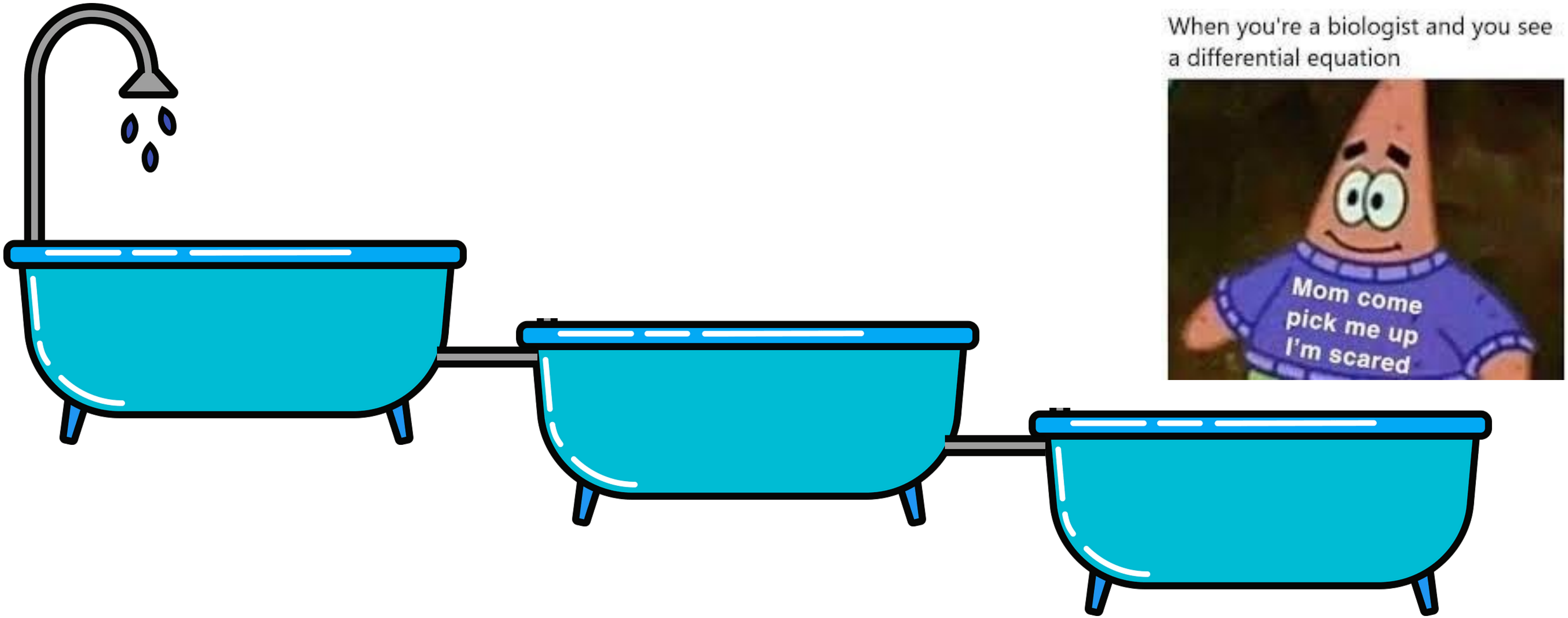
These are the "compartments"



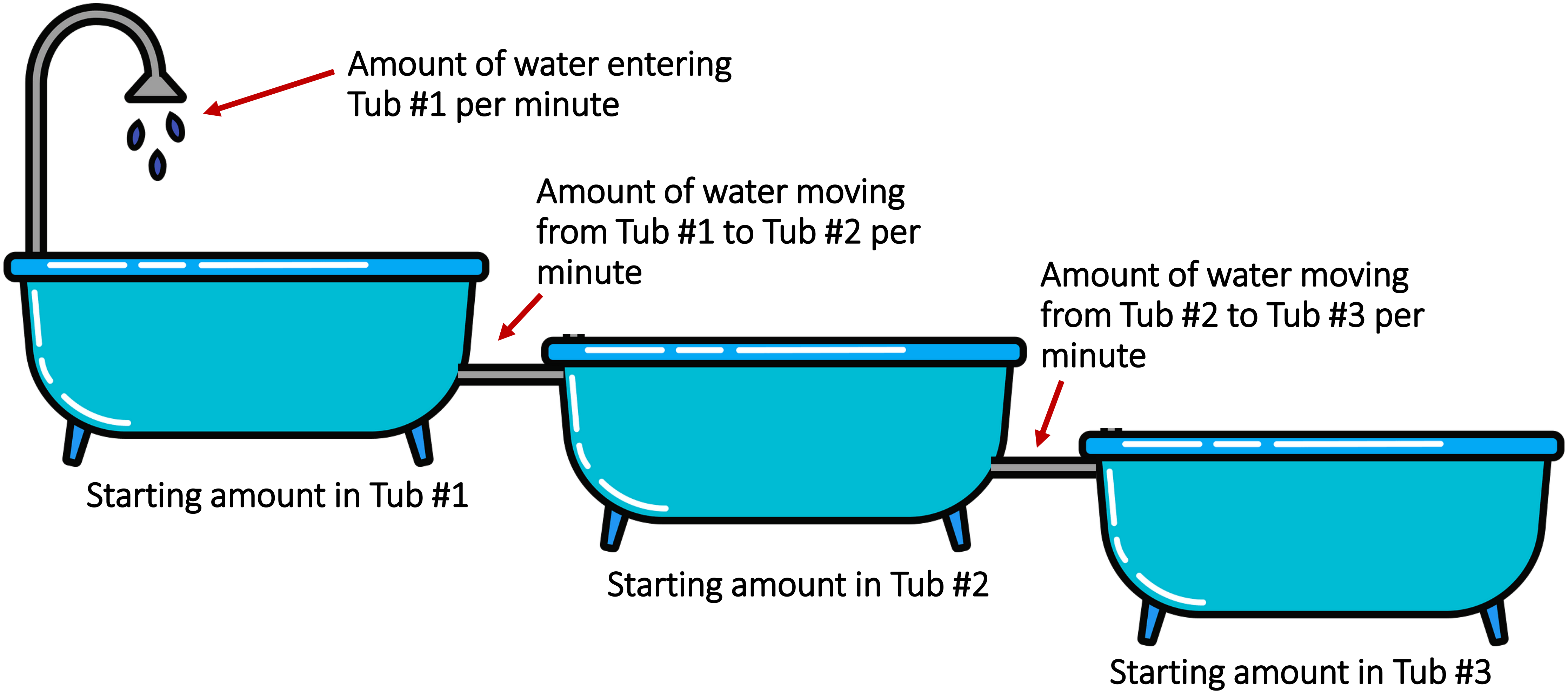
These are the “rates”



The goal is to predict the water level in each bathtub over time



We can use differential equations to model this system

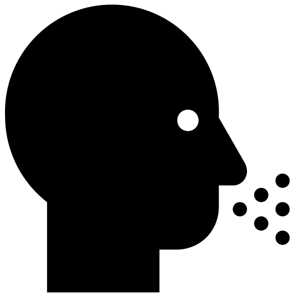


We can use differential equations to model this system

The Foundational Compartmental Model in Epidemiology
SIR

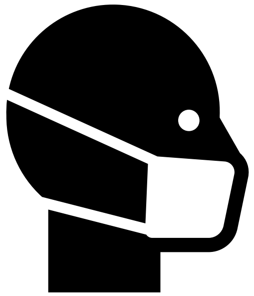
SUSCEPTIBLE

- ❑ Able to become infected
- ❑ Unable to infect others



INFECTIOUS

- ❑ Unable to become infected
- ❑ Able to infect others



RECOVERED

- ❑ Unable to become infected
- ❑ Unable to infect others

Parameters
β – Transmission rate
γ - Recovery rate
N – Total #
I – Infected #
S – Susceptible #
T – Time unit

Basic SIR Model

$$\frac{dS}{dt} = -\frac{\beta I}{N} S \qquad \frac{dI}{dt} = \frac{\beta IS}{N} - \gamma I \qquad \frac{dR}{dt} = \gamma I$$

Parameters

β – Transmission rate

γ - Recovery rate

N – Total #

I – Infected #

S – Susceptible #

T – Time unit

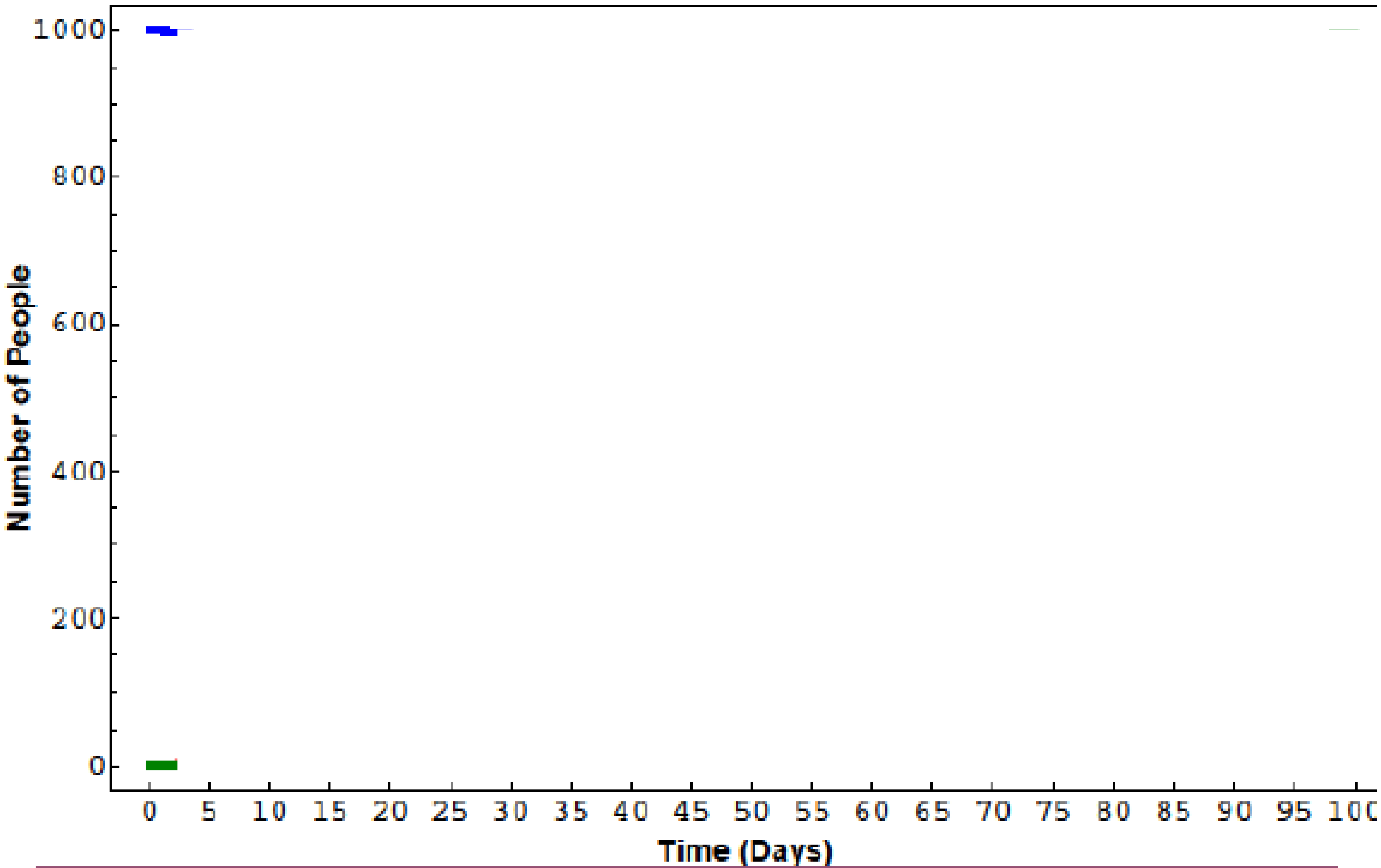
Basic SIR Model

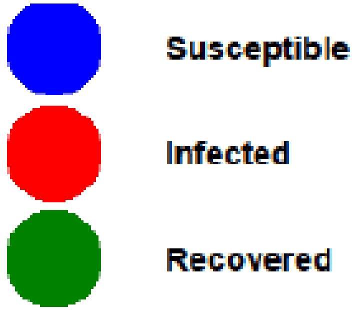
Hint: you've seen one of these parameters before…

$$R_0 = \frac{\beta}{\gamma}$$

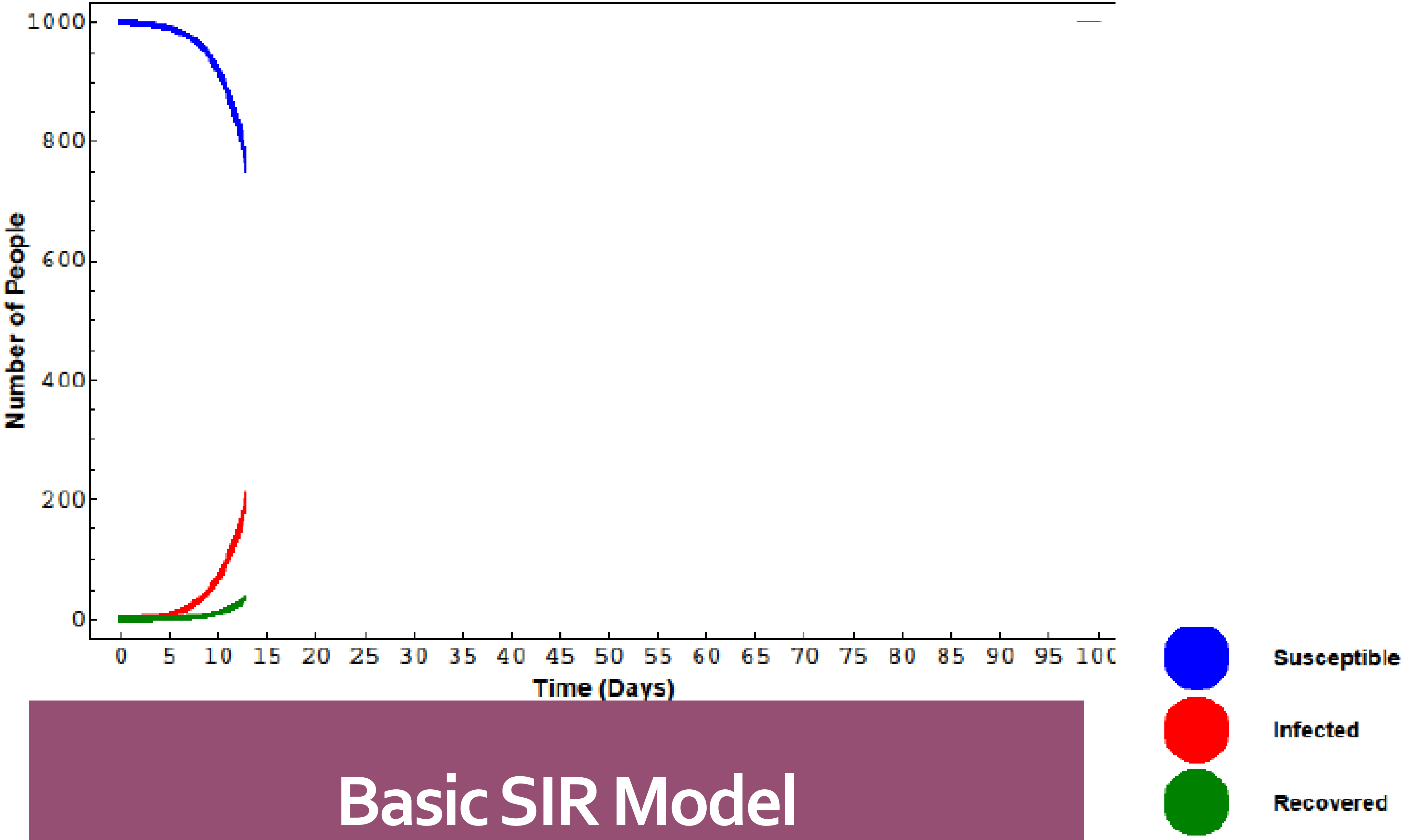
Pause.
Reflect.
Ask questions.

What’s your favorite disease?
Could you study it with an SIR model?

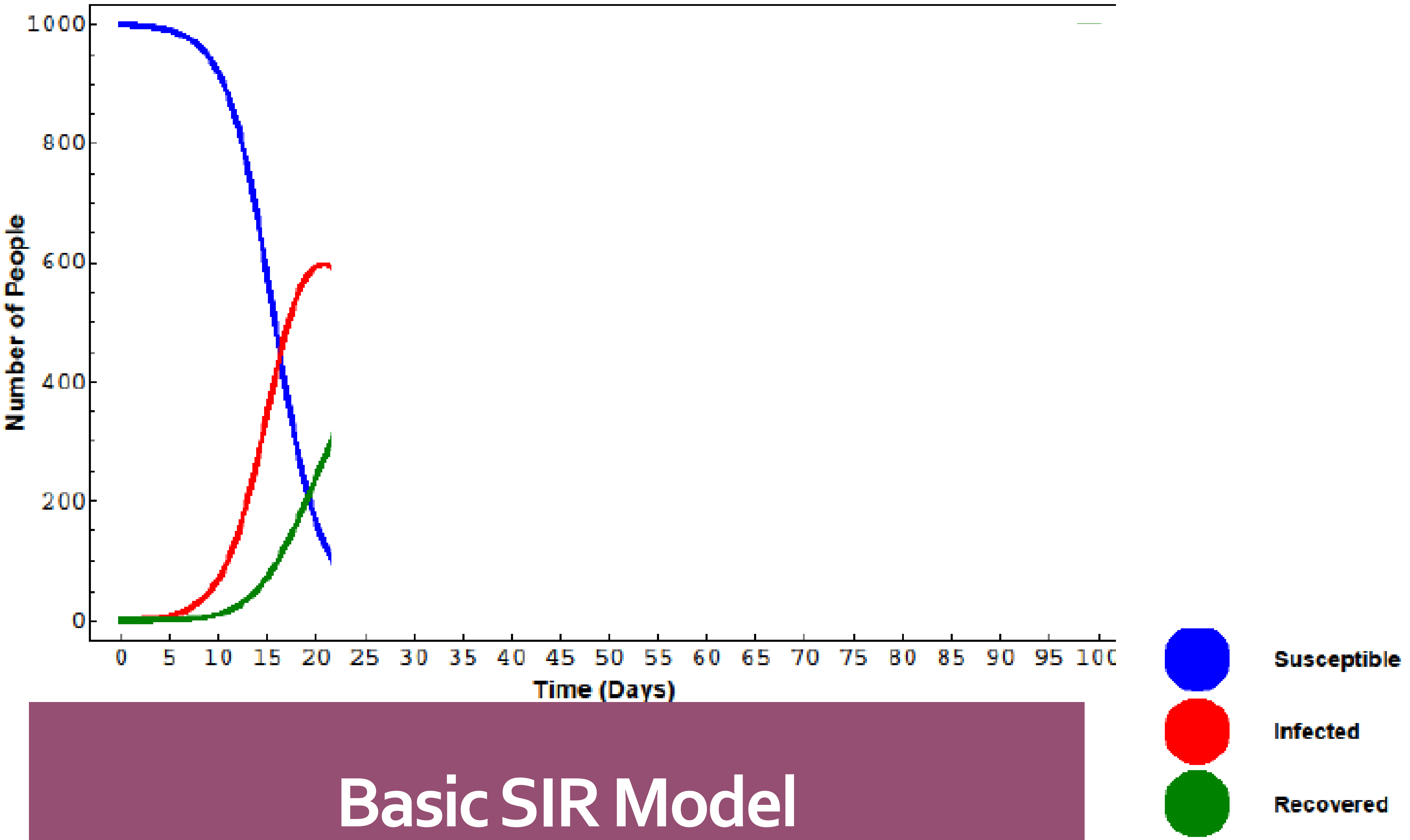




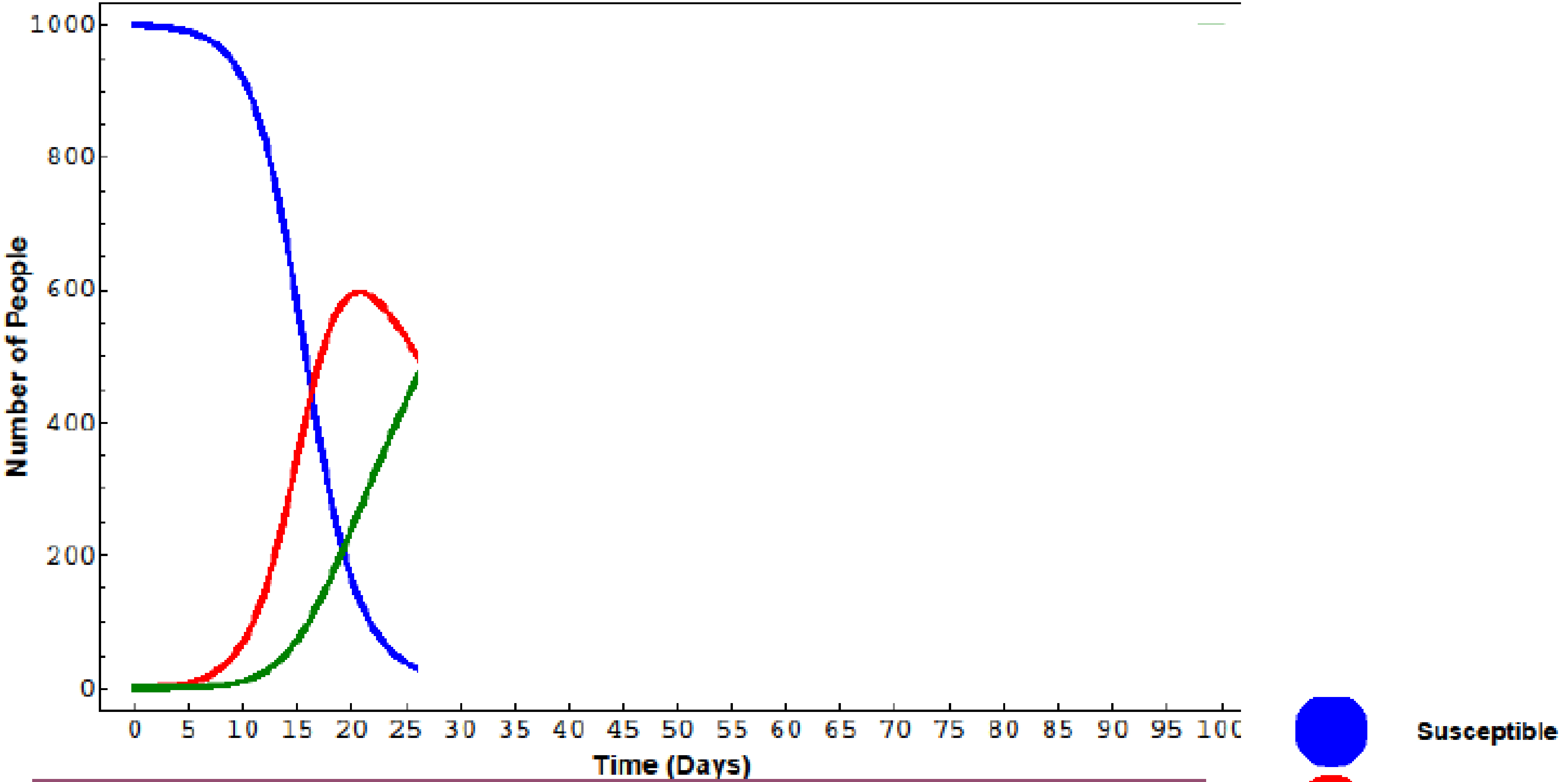
Basic SIR Model



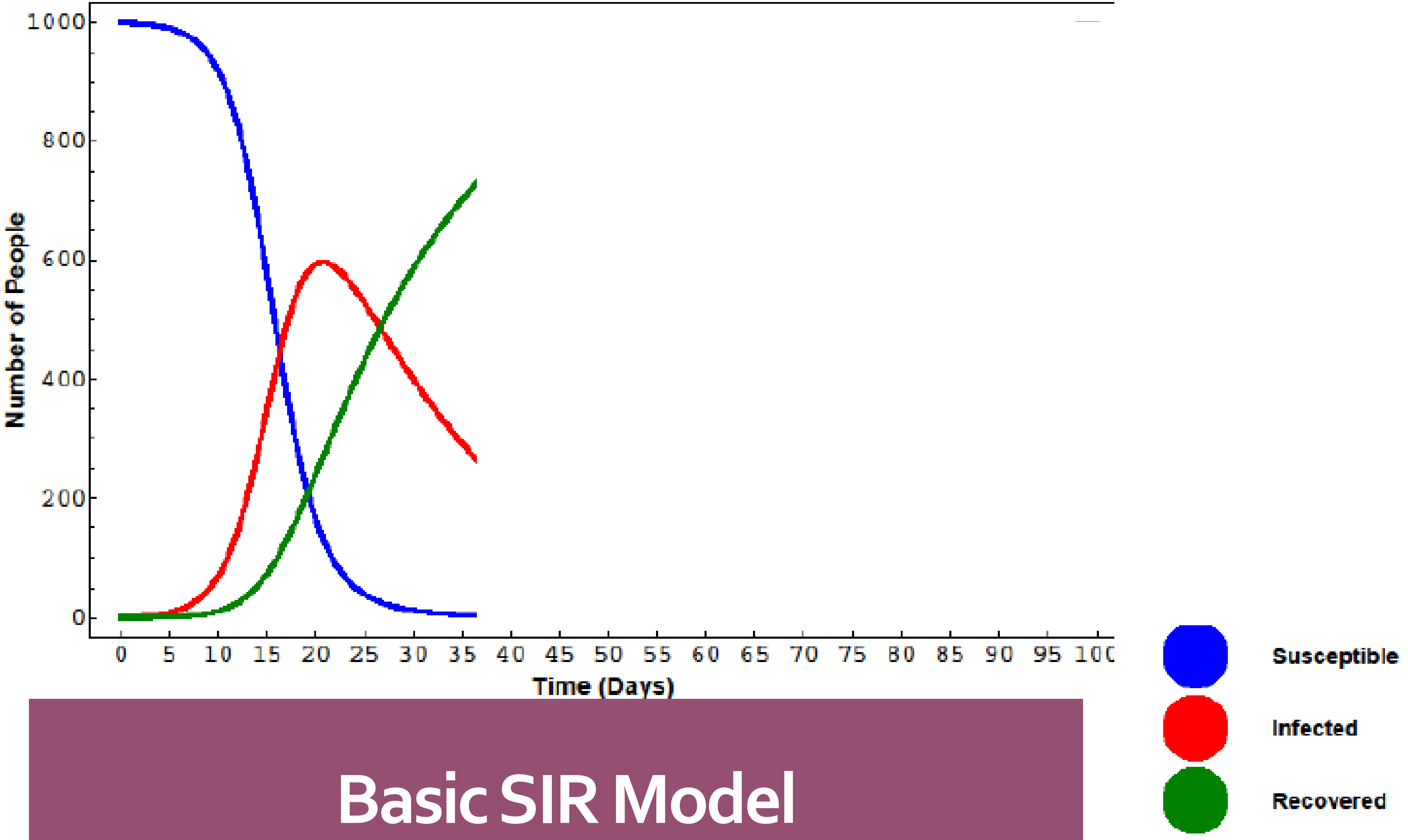
Basic SIR Model



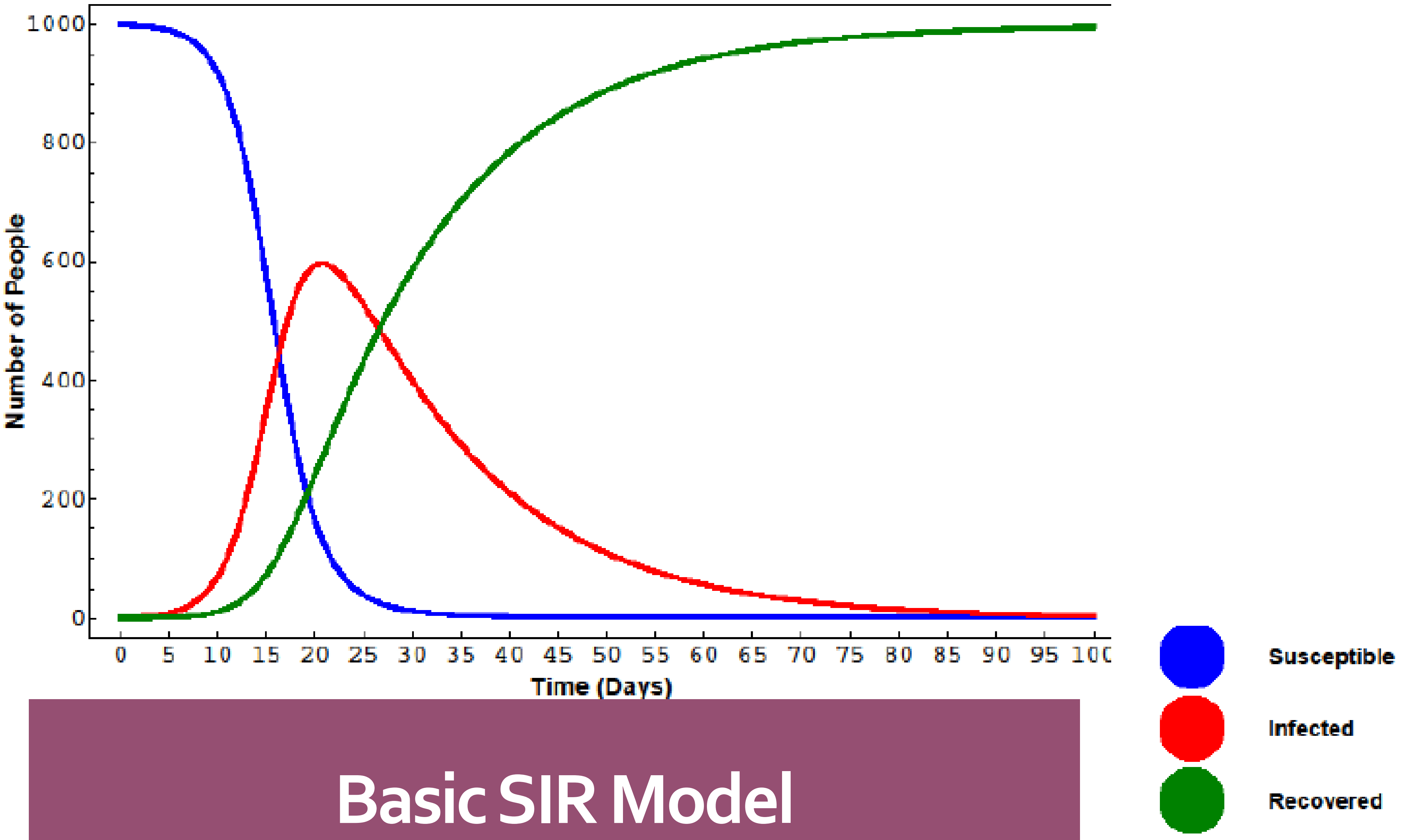
Basic SIR Model



Basic SIR Model



Basic SIR Model



Basic SIR Model

Pause.
Reflect.
Ask questions.

Think about the parameters in an SIR model. How might the model predictions change if you increased or decreased it?

Parameters

β – Transmission rate

γ - Recovery rate

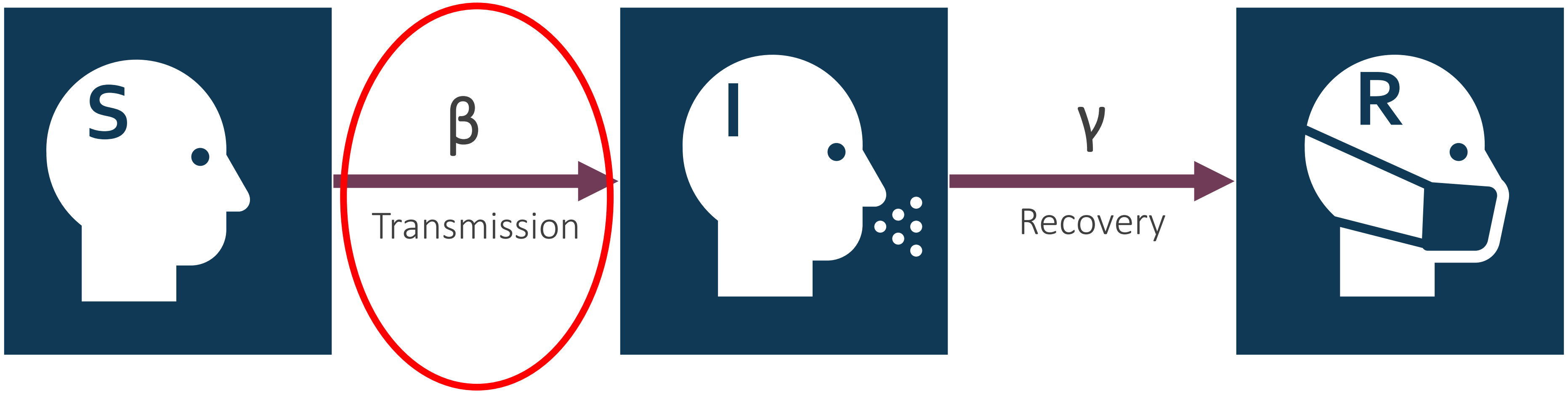
N – Total #

I – Infected #

S – Susceptible #

T – Time unit

Stay-at-home orders… what does it change?



Parameters

β – Transmission rate

γ - Recovery rate

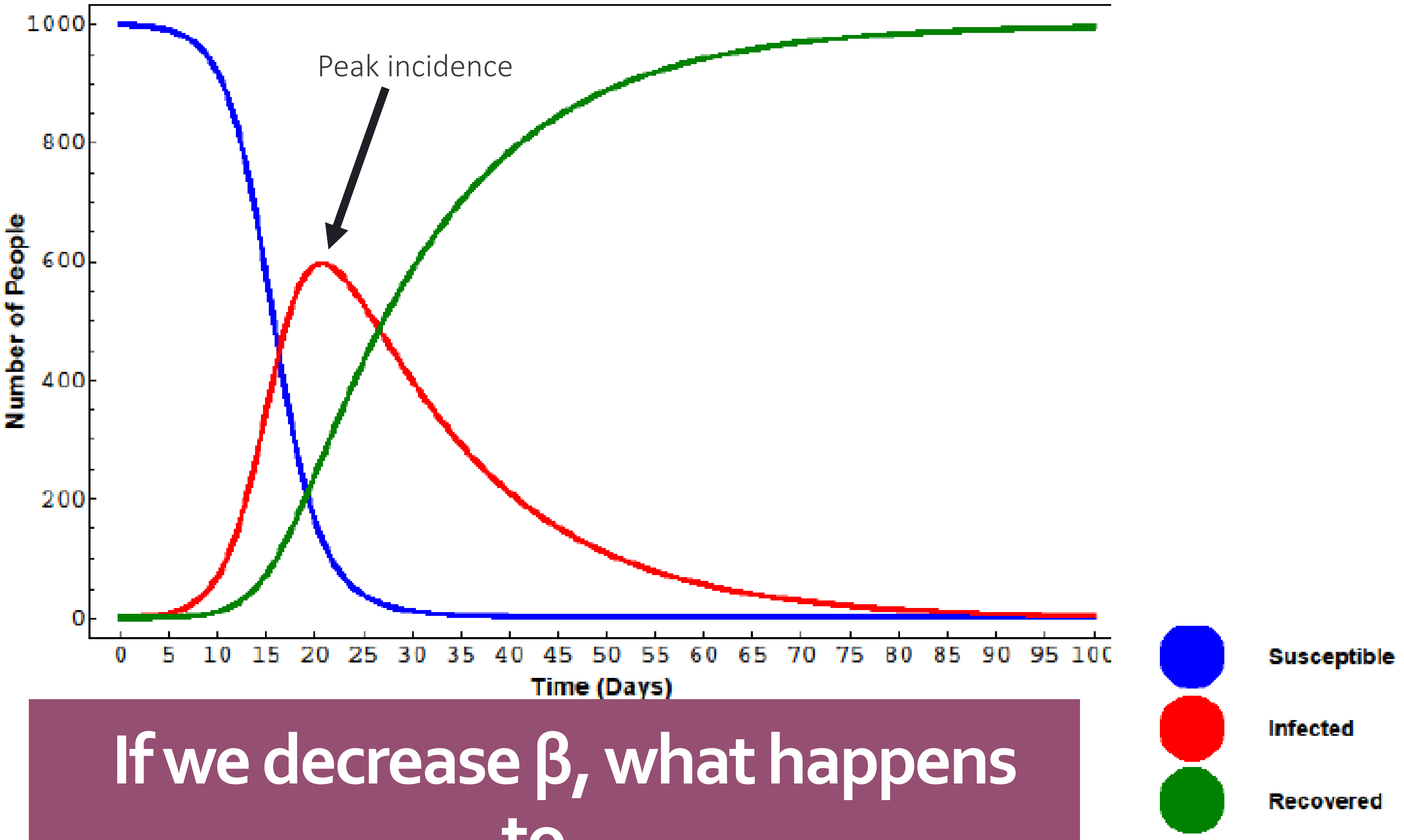
N – Total #

I – Infected #

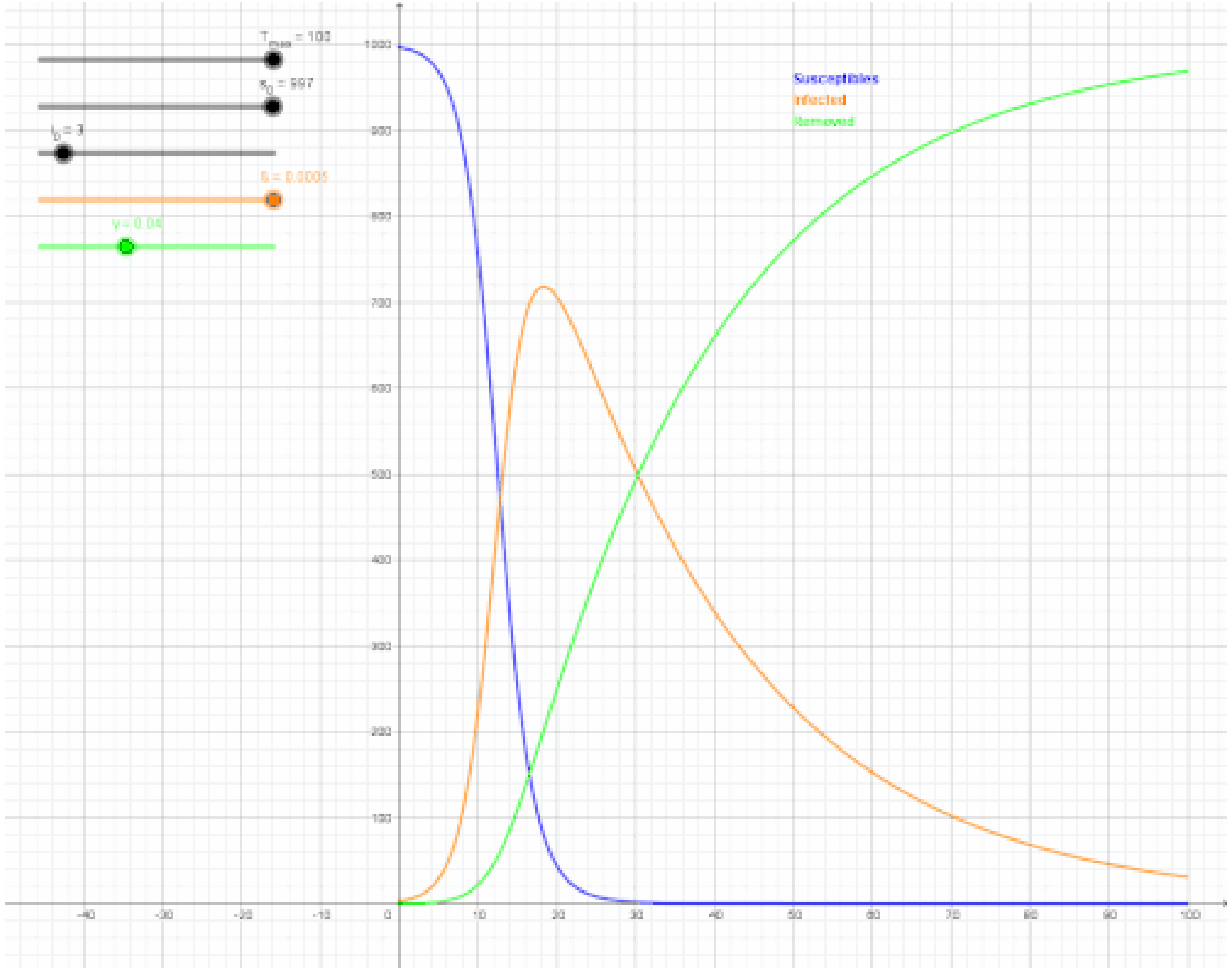
S – Susceptible #

T – Time unit

Decreases transmission rate!



If we decrease β, what happens to…



“Flattening the Curve”

Expanding on Basic SIR Model

S Susceptible

E *EXPOSED*

Latency period!

I Infectious

R Recovered

Loss of immunity!

S *SUSCEPTIBLE AGAIN*

Any questions?

Let's take a short bio break and then practice with our own SIR models!